

ML · CLASSIFICATION PROJECT

Will the Bank Say **YES** or **NO**?

Predicting Loan Approvals with Machine Learning

...

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Python · Scikit-learn · Pandas · Seaborn





70.2%

of applicants get REJECTED

Dataset: 1,000 real applicants

The Problem



Banks manually review thousands of loan applications



Slow, inconsistent, and prone to human bias



ML can automate approval decisions instantly

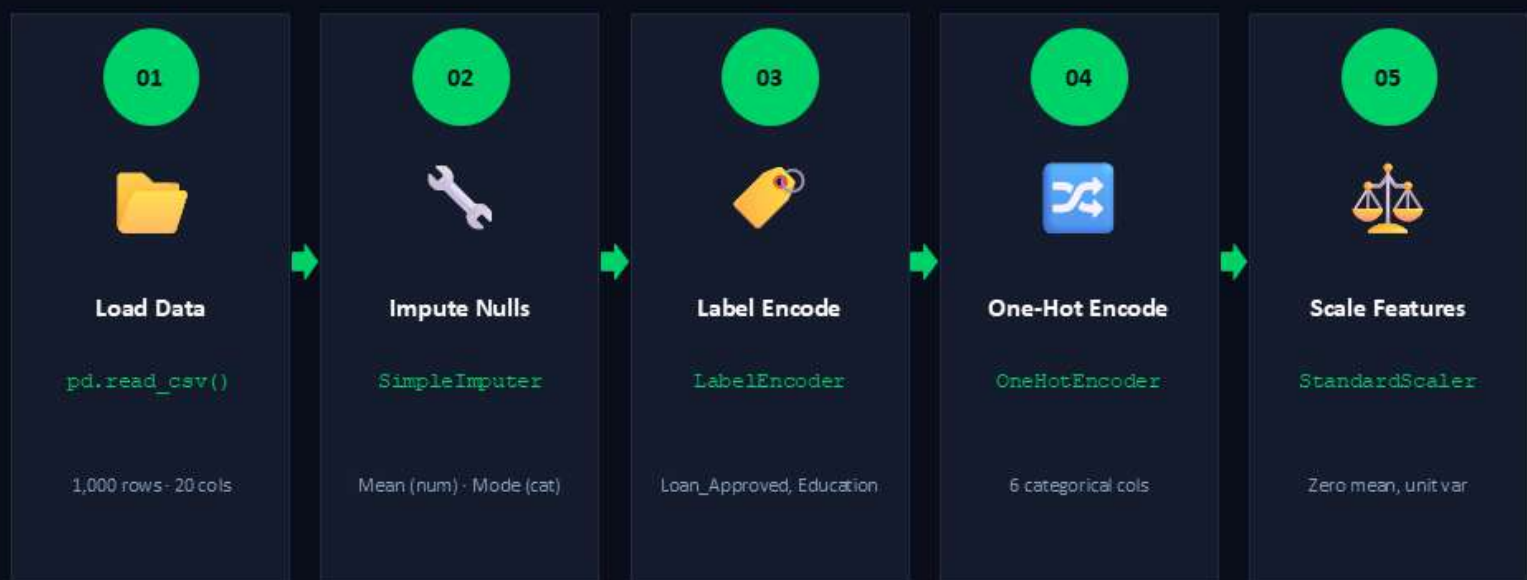


Goal: Train an ML model that predicts loan approval

Dataset at a Glance



Preprocessing Pipeline



⚡ Feature Engineering: `DTI_Ratio2` and `Credit_Score2` added to capture non-linear patterns

3 Models. 1 Winner.

All trained on same 80/20 split · StandardScaler applied



Logistic Regression

Linear · Probabilistic

- Best for binary classification
- Fast & Interpretable
- Handles scaled features well



K-Nearest Neighbors

Distance-based · k=5

- No model training step
- Sensitive to feature scale
- Affected by noisy data



Naive Bayes

Gaussian · Probabilistic

- Assumes feature independence
- Extremely fast training
- Works well on small datasets

Results

🏆 Best Model: Logistic Regression · Highest Accuracy & F1 across both rounds

Round 1 — Baseline

Model	Accuracy	F1
Logistic Reg	~82%	~0.75
KNN	~79%	~0.71
Naive Bayes	~75%	~0.66



improved

Round 2 — After Feature Engineering

Model	Accuracy	F1
Logistic Reg	~84%	~0.78
KNN	~81%	~0.74
Naive Bayes	~77%	~0.69

⚡ Feature Eng. boosted all models

📊 High precision = fewer false approvals

🔗 Credit Score & DTI are top predictors

Key Learnings



Encode everything

Label Encode targets · One-Hot for multi-category columns



Impute before you split

Fit imputer on train only to prevent data leakage



Scale for distance models

StandardScaler is essential for KNN & Logistic Regression



Engineer smart features

DTI² & Credit Score² captured non-linear patterns, boosted accuracy



Accuracy isn't enough

Imbalanced data (70/30 split) — always check F1 + Confusion Matrix



Simple models still win

Logistic Regression outperformed KNN and Naive Bayes consistently

Thanks for reading!



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